

Binary, Octal & Hexadecimal Operations

Week 5

Topics:

Binary Arithmetic Operations

Octal Arithmetic Operations

Hexadecimal Arithmetic Operations

Objectives

At the end of the class, you should be able to perform mathematical operations such as

- Addition
- Subtraction
- Multiplication
- Division

involving binary, octal and hexadecimal numbers

Arithmetic Operations of Decimal Numbers

$$\begin{array}{r} 1 \\ 12 \\ + 29 \\ \hline 41 \end{array} \qquad \begin{array}{r} 41 \\ - 29 \\ \hline 12 \end{array}$$

Binary Addition

Addition of Binary Numbers

- $0 + 0 = 0$ (or 0_{10})
- $0 + 1 = 1$ (or 1_{10})
- $1 + 0 = 1$ (or 1_{10})
- $1 + 1 = 10$ (or 2_{10})

DEC	BIN	OCT	HEX
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Binary Addition

$$(11010)_2 + (1100)_2$$

$$\begin{array}{r} \boxed{1} \boxed{1} \leftarrow \text{Carry} \\ 11010 \\ + \quad 1100 \\ \hline 100110 \end{array}$$

Binary Subtraction

The following are the 4 basic subtractions of the binary system:

- $0 - 0 = 0$
- $0 - 1 = -1$
- $1 - 0 = 1$
- $1 - 1 = 0$

Among these subtractions, if the upper digit of 0 is 1 in $0 - 1 = -1$, a "borrow" is performed.

$$\begin{array}{r} \heartsuit \quad \leftarrow \text{Borrow} \\ 10 \\ - 1 \\ \hline 1 \end{array}$$

Binary Subtraction

Example

 $(10011)_2 - (1001)_2$

$$\begin{array}{r} \heartsuit \quad \leftarrow \text{Borrow} \\ 10011 \\ - 1001 \\ \hline 1010 \end{array}$$

Binary Multiplication

Multiplication

$$0 \times 0 = 0$$

$$0 \times 1 = 0$$

$$1 \times 0 = 0$$

$$1 \times 1 = 1$$

Example:

$$\begin{array}{r} 101 \\ * 11 \\ \hline 101 \\ 101 \\ \hline 1111 \end{array}$$

DEC	BIN	OCT	HEX
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Binary Division

Binary Division

$$\begin{array}{r} 10011 \text{ r } 10 \\ 11 \overline{) 111011} \\ \underline{-11} \\ 101 \\ \underline{-11} \\ 101 \\ 11 \\ \underline{11} \\ 10 \end{array}$$

Octal Numbers Arithmetic Operations

The method to perform arithmetic operations for octal number is similar to the method in performing arithmetic operations for decimal, binary and hexadecimal

Octal Addition Table

+	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	10
2	2	3	4	5	6	7	10	11
3	3	4	5	6	7	10	11	12
4	4	5	6	7	10	11	12	13
5	5	6	7	10	11	12	13	14
6	6	7	10	11	12	13	14	15
7	7	10	11	12	13	14	15	16

Source: <https://www.geeksforgeeks.org/arithmetic-operations-of-octal-numbers/>

Octal Numbers Arithmetic Operations

Example 1: $(7)_8 + 2_8 = 11$

Example 2:
$$\begin{array}{r} (16)_8 \\ + (23)_8 \\ \hline (41)_8 \end{array}$$

Example 3: Get the sum of the ff
Octal numbers

$$\begin{array}{r} 7602 \\ + 5711 \\ \hline \end{array}$$

DEC	BIN	OCT	HEX
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Octal Addition and Subtraction

$$\begin{array}{r} (41)_8 \\ + (11)_8 \\ \hline (52)_8 \end{array}$$

$$\begin{array}{r} (41)_8 \\ + (37)_8 \\ \hline (100)_8 \end{array}$$

$$\begin{array}{r} (41)_8 \\ - (37)_8 \\ \hline (2)_8 \end{array}$$

$$\begin{array}{r} 43 \\ + 25 \\ \hline \end{array}$$

$$\begin{array}{r} 76 \\ + 23 \\ \hline \end{array}$$

$$\begin{array}{r} 55 \\ + 44 \\ \hline \end{array}$$

$$\begin{array}{r} 52 \\ - 37 \\ \hline \end{array}$$

$$\begin{array}{r} 76 \\ - 32 \\ \hline \end{array}$$

$$\begin{array}{r} 55 \\ - 23 \\ \hline \end{array}$$

Octal Multiplication Table

x	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	10	12	14	16
3	0	3	6	11	14	17	22	25
4	0	4	10	14	20	24	30	34
5	0	5	12	17	24	31	36	43
6	0	6	14	22	30	36	44	52
7	0	7	16	25	34	43	52	61

Example 1:

$$\begin{array}{r}
 (41)_8 \\
 \times (11)_8 \\
 \hline
 41 \\
 41 \\
 \hline
 451_8
 \end{array}$$

Other approach

Example 2:

$$\begin{array}{ccc}
 (47)_8 & \rightarrow \text{Decimal} & 39 \\
 \times (35)_8 & \rightarrow \text{Decimal} & \times 29 \\
 \hline
 (2153)_8 & \leftarrow & (1131)_{10}
 \end{array}$$

Octal Multiplication

$$\begin{array}{r} (41)_8 \\ \times (11)_8 \\ \hline \end{array}$$

$$\begin{array}{r} (41)_8 \\ \times (37)_8 \\ \hline \end{array}$$

$$\begin{array}{r} (41)_8 \\ \times (37)_8 \\ \hline \end{array}$$

$$\begin{array}{r} 43 \\ \times 5 \\ \hline \end{array}$$

$$\begin{array}{r} 6 \\ \times 23 \\ \hline \end{array}$$

$$\begin{array}{r} 55 \\ \times 4 \\ \hline \end{array}$$

Octal Division

Example:

$$O(1160) / O(20) = O(47)$$

Try the following:

1. $O(1053) / O(17) = O(??)$

2. $O(600) / D(16) = O(??)$

Hexadecimal arithmetic operations

Addition and subtraction

✓ Addition

- Perform addition starting at the lowest significant digit
- A carry to the upper digit is performed when the result is higher than 16

✓ Subtraction

- Performed starting at the lowest significant digit
- A borrow from the upper digit is performed when the result is negative

Hexadecimal Addition Table

+	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
1	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	10
2	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	10	11
3	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	10	11	12
4	05	06	07	08	09	0A	0B	0C	0D	0E	0F	10	11	12	13
5	06	07	08	09	0A	0B	0C	0D	0E	0F	10	11	12	13	14
6	07	08	09	0A	0B	0C	0D	0E	0F	10	11	12	13	14	15
7	08	09	0A	0B	0C	0D	0E	0F	10	11	12	13	14	15	16
8	09	0A	0B	0C	0D	0E	0F	10	11	12	13	14	15	16	17
9	0A	0B	0C	0D	0E	0F	10	11	12	13	14	15	16	17	18
A	0B	0C	0D	0E	0F	10	11	12	13	14	15	16	17	18	19
B	0C	0D	0E	0F	10	11	12	13	14	15	16	17	18	19	1A
C	0D	0E	0F	10	11	12	13	14	15	16	17	18	19	1A	1B
D	0E	0F	10	11	12	13	14	15	16	17	18	19	1A	1B	1C
E	0F	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D
F	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E

<http://www.pcc-services.com/binhex.html>

Hexadecimal arithmetic operations

$$(A8D)_{16} + (B17)_{16}$$

$$\begin{array}{r} \boxed{1} \boxed{1} \leftarrow \text{Carry} \\ A8D \\ + B17 \\ \hline 15A4 \end{array}$$

$$\left[\begin{array}{r} 1 \quad 1 \\ 10 \quad 8 \quad 13 \\ + 11 \quad 1 \quad 7 \\ \hline 21 \quad 9 \quad 20 \\ 1 \quad 5 \quad A \quad 4 \end{array} \right]$$

$D + 7 = (\text{In the decimal system: } 13 + 7 = 20) = 16 \text{ (carried 1)} + 4$
The sum of the first column is 4 and 1 is carried to the second column.

$1 + 8 + 1 = (\text{In the decimal system: } 10) = A$

↑ Carried from the first column

$A + B = (\text{In the decimal system: } 10 + 11 = 21) = 16 \text{ (carried 1)} + 5$

The sum of the third column is 5 and 1 is carried to the fourth column.

Result = $(15A4)_{16}$

Hexadecimal arithmetic operations

$$\begin{array}{r} \heartsuit \leftarrow \text{Borrow} \\ 6D3 \\ - 174 \\ \hline 55F \end{array}$$

$$\left[\begin{array}{r} \heartsuit 16 \\ 6 \quad 13 \quad 3 \\ - 1 \quad 7 \quad 4 \\ \hline 5 \quad 5 \quad 15 \end{array} \right]$$

First column from right

- Since $3 - 4 = -1$, a borrow is performed from D in the second digit (D becomes C).
- $16 \text{ (borrowed 1)} + 3 - 4 = F$ (In the decimal system: $19 - 4 = 15$)

Second column from right

- $C - 7 = 5$ (In the decimal system: $12 - 7 = 5$)

Third column

- $6 - 1 = 5$

The result is $(55F)_{16}$.

Hexadecimal Multiplication Table															
*	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
1	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
2	02	04	06	08	0A	0C	0E	10	12	14	16	18	1A	1C	1E
3	03	06	09	0C	0F	12	15	18	1B	1E	21	24	27	2A	2D
4	04	08	0C	10	14	18	1C	20	24	28	2C	30	34	38	3C
5	05	0A	0F	14	19	1E	23	28	2D	32	37	3C	41	46	4B
6	06	0C	12	18	1E	24	2A	30	36	3C	42	48	4E	54	5A
7	07	0E	15	1C	23	2A	31	38	3F	46	4D	54	5B	62	69
8	08	10	18	20	28	30	38	40	48	50	58	60	68	70	78
9	09	12	1B	24	2D	36	3F	48	51	5A	63	6C	75	7E	87
A	0A	14	1E	28	32	3C	46	50	5A	64	6E	78	82	8C	96
B	0B	16	21	2C	37	42	4D	58	63	6E	79	84	8F	9A	A5
C	0C	18	24	30	3C	48	54	60	6C	78	84	90	9C	A8	B4
D	0D	1A	27	34	41	4E	5B	68	75	82	8F	9C	A9	B6	C3
E	0E	1C	2A	38	46	54	62	70	7E	8C	9A	A8	B6	C4	D2
F	0F	1E	2D	3C	4B	5A	69	78	87	96	A5	B4	C3	D2	E1

<http://www.pcc-services.com/binhex.html>

Hexadecimal Multiplication

Example 1:

$$\begin{array}{r} 9_{16} \\ \times 9_{16} \\ \hline \end{array}$$

81₁₀ → (51)₁₆

How?

$$\begin{array}{r|l} 16 & 81 \\ 16 & \underline{5} \quad 1 \\ & \underline{0} \quad 5 \end{array}$$

Example 2:

$$\begin{array}{r} 5A3 \\ \times 1B \\ \hline \end{array}$$

	5	10	3
	x	1	11

	55	110	33
5	10	3	

5	65	113	33

In Hexa: 5 **41** **71** **21**

In Hexa: 9 8 3 1

Hexadecimal Division

Example 1: $H(51)/H(9) = (5 \cdot 16 + 1)/9 = (81)_{10}/(9)_{10} = (9)_{10} \Rightarrow \text{Hexa} = H(9)$

Example 2: $H(9830)/H(A) = (38960)_{10}/(10)_{10} = 3896_{10} = (F38)_{16}$

Other approach

Example 2:

5A3	→ Decimal	1443
x 1B	→ Decimal	x 27
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(9831) ₁₆	←	(38961) ₁₀

Exercises

Compute the following

- a) $11011_2 + 1111_2$
- b) $33_8 + 17_8$
- c) $1B_{16} + F_{16}$
- d) $50_{10} - 22_{10}$
- e) $110010_2 - 10110_2$
- f) $62_8 - 26_8$
- g) $32_{16} - 16_{16}$
- h) $32_{16} * 16_{16}$
- i) $106_8 / 5_8$

Exercises

- Compute the following

- a) $27_{10} + 15_{10}$
- b) $11011_2 + 1111_2$
- c) $33_8 + 17_8$
- d) $1B_{16} + F_{16}$

- Compute the following

- a) $50_{10} - 22_{10}$
- b) $110010_2 - 10110_2$
- c) $62_8 - 26_8$
- d) $32_{16} - 16_{16}$



References:

1. Decimal Number System. <http://www.britannica.com/topic/decimal-number-system>
2. <http://www.pcc-services.com/binhex.html>
3. <http://www.math.grin.edu/~rebelsky/Courses/152/97F/Readings/student-binary>
4. <http://www.doc.ic.ac.uk/~eedwards/compsys/arithmetic/>
5. <https://www.geeksforgeeks.org/arithmetic-operations-of-octal-numbers/>

Thank you!